

Evaluating Interactive Technology with Children

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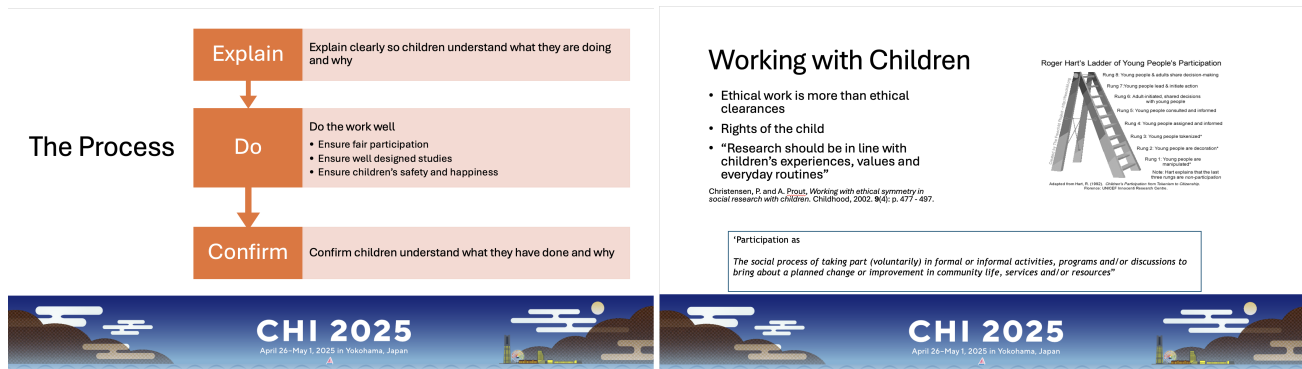


Figure 1: Images of Course Materials

Abstract

While evaluating technology with adults is well understood, evaluating interactive technology with child users has received far less attention and raises a range of unusual and unexpected challenges. With more children than ever before using interactive technology on a daily basis, this course, for practitioners and researchers, aims to provide a succinct and useful introduction to evaluating technology with children. The course begins with the developmental and ethical challenges of working with children, then covers a range of foundational evaluation techniques, along with how techniques are used and how results are reported both in publications and to child audiences.

CCS Concepts

• **Human-centered computing** → HCI theory, concepts and models.

Keywords

Evaluation, Usability, UX, User Studies, Children

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1 Benefits

Few courses cover the specifics of working with children in HCI contexts. This course fills a gap for many students and practitioners by delivering understandings of how to successfully evaluate systems with children and draws on the best practice from industry and academia. It would benefit anyone wishing to understand technology use in the context of child users. Students attending will:

- Be able to critically consider the purpose and value of evaluation with children.
- Understand the range of evaluation techniques available and how they are used.
- Be able to choose an appropriate technique for a specific context.
- Understand how the results of an evaluation should be presented in a research paper.
- Appreciate, and learn how to address, the unique difficulties and challenges inherent in working with child users.
- Understand the importance of inclusion when working with children.

As well as experiencing a range of practical activities with a set of experienced instructors, participants will hear about three quite different evaluation studies with children ranging from tots to teens, they will examine their own choices in regards to how children participate and will be given support after the course via an online portal where concerns and ideas can be shared.

2 Intended Audience

This course is valuable to students and practitioners within the CHI community who focus on children as a user group in their work. The specific intended audience for this course is those who have a strong interest in the successful evaluation of technology products for children.

3 Prerequisites

There are no prerequisites to this course aside from a general knowledge of HCI as could be gathered from an undergraduate module / course on the same or from work in the UX industry.

4 Content

The content is broken into six themes:

- (1) **Landscape** Considering the product, its purpose, its potential users and the goal of conducting an evaluation, this intro section shows the huge array of different possibilities for evaluation. *Learning Outcome:* Be able to critically consider the purpose and value of evaluation with children.
- (2) **Techniques** Referencing accompanying notes (where much more detail will be found) we look at a sample of evaluation techniques that address different positions in our landscape - these include playtesting, evaluating fun, discovering usability flaws, evaluating long-term use, diary studies and UX surveys [12] [2] [13], [11]. *Learning Outcome:* Understand the range of evaluation techniques available and how they are used.
- (3) **Choosing Techniques** Through consideration and discussion of a set of scenarios we look at how techniques can be chosen to answer a question. *Learning Outcome:* Be able to choose an appropriate technique for a specific context.
- (4) **Reporting** In this theme we discuss how we report what we find, through conducting an evaluation, to a range of audiences [7]. *Learning Outcome:* Understand different ways that the results of an evaluation can be presented.
- (5) **Challenges** From their extensive experiences, the organizers will share key challenges they have faced when conducting evaluation work with children and will offer some solutions. *Learning Outcome:* Appreciate the unique difficulties and challenges inherent in working with child users and how to address them.
- (6) **Better Inclusion of Children** This theme considers ways to ensure full inclusion of children in evaluation studies and introduces core concepts around children's rights and the ethics of working with children [9]. *Learning Outcome:* Understand the importance of inclusion when working with children.

5 Practical Work

There are seven practical exercises within this course:

- (1) **Finding out about one another** This has proven useful in other courses as it also encourages networking of participants both within and after the course. Participants will briefly describe their experience of working with children and their rationale for coming to the course. They will also

be encouraged to state what specific questions they want answering.

- (2) **Rating scenarios** Participants will be presented with scenarios of evaluations with children and will then rate and discuss them to help identify 'good' and 'bad' examples. These scenarios will include a user study with a commercial company with very young children, an at home study with a public service company with children with disabilities, and a research study exploring interface design with school children.
- (3) **Considering sampling of children** Using a specially designed card deck participants will arrange these in order against different criteria in a group activity. This activity will assist in the discussion of representative samples, convenience samples and targeted samples; with discussion on the size of samples needed to show different outcomes.
- (4) **Planning** Participants will examine planning sheets in small groups to determine whether these are user studies, usability studies or UX studies. The aim with this activity is to help participants clarify the intended outcomes of a study in order to better explain outcomes to children and others.
- (5) **Analysing Data** Participants will work individually to look at survey data from children and interpret what the children produced. This activity highlights different survey designs and shows how analysis differs according to instrument design.
- (6) **Fun Toolkit** As a group we will work through the the Fun Toolkit instrument and explore how to report the data it produces.
- (7) **Reporting Back** Working from examples of work that is already published the participants will work in groups to design novel 'reporting back' activities for children.

6 Instructor Background

Prof Janet C Read is a full professor with over 20 years experience of working with children in HCI during which time she has examined over 35 PhD students across 13 different countries. She was a co-author of the book 'Evaluating Interactive Technology with Children' [3] and has recently contributed a chapter on Ethics in HCI to the Foundations in HCI series [8]. She has organised several courses at CHI e.g. [5], [6] and elsewhere where working with children has been a focus. Her Fun Toolkit [10] is the most often used evaluation method for children in HCI, and has been adopted by industry.

Dr Matt Horton is a senior lecturer in UX with a long trajectory of research with children in HCI. He was instrumental in organising two IDC conferences in Manchester and Preston, the British HCI conference in Southport UK, and has co-delivered courses on Child Computer Interaction at CHI e.g. [4], [6]. He is the main author of the paper describing the MESS day research protocol [1].

Dr Daniel Fitton's research focus is on HCI and UX issues within the context of child users with an emphasis on evaluation and design techniques for this group. He currently has over 100 peer-reviewed publication in the field of HCI and has organized many successful workshops at past CHI conferences, in addition to having

previously been full-papers chair at the ACM Interaction Design and Children (IDC) conference.

7 Resources

Course materials and details of the activities on the day will be provided at www.chici.org/EvalCourse2025

8 Accessibility

To support inclusion we will ensure materials and information about activities are made available to attendees in advance of the workshop. Attendees are encouraged to contact the course organizers to discuss accessibility arrangement where appropriate.

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